

AI Sales Agent: Solution Documentation

Task Overview

This document summarizes the implementation of AI Sales Agent, providing details on the approach, functionality, and challenges faced during development.

Approach to AI Sales Agent

Technology Stack

- **Frontend:** Streamlit for interactive UI
- **Backend:** Python for data processing and business logic
- **AI Integration:** Google Gemini API for natural language conversations
- **Configuration:** dotenv for environment management

Implementation Strategy

1. **Modular Design:** Separated the application into distinct components:
 - `main.py`: UI and application flow
 - `sales_agent.py`: AI conversation logic
 - `product_catalog.py`: Product data and recommendation engine
 - `config.py`: API configuration
2. **Conversational Design:**
 - Created a chat-based interface
 - Implemented context preservation between messages
 - Designed prompts to guide AI responses toward sales assistance
3. **Product Intelligence:**
 - Built a structured product catalog with categories
 - Implemented category detection in natural language
 - Added alias mapping for common terms (e.g., "phone" → "smartphone")

Functionality Demonstration

Scenario 1: Product Category Exploration

When a user inquires about a product category, the system detects the category and offers exploration options:

AI Sales Agent

Ask me anything about our products, recommendations, purchases, or support!



I need to buy gaming laptop.



Okay! To recommend the best gaming laptop for you, I need a little more information. Could you tell me about:

- **Your budget:** How much are you looking to spend?
- **What games do you want to play?** (This helps determine the necessary specs)
- **What size screen are you looking for?** (e.g., 15-inch, 17-inch)
- **Are there any specific features you need?** (e.g., long battery life, specific ports, lightweight for travel)

Once I have this information, I can suggest some excellent options from our catalog!

Product Categories:

[View Laptop Products](#)

[Reset Chat](#)

Type your message here...



The system identifies product categories mentioned in user messages and displays category buttons for quick navigation.

Scenario 2: Product Recommendations

The AI agent provides personalized product recommendations based on user preferences:



400000 LKR Shooting games 15inch Long battery life



Okay, great! Based on your requirements of a 400,000 LKR budget, playing shooting games, a 15-inch screen, and long battery life, I recommend the **ASUS ROG Zephyrus G15**.

While hitting the long battery life mark in a gaming laptop can be tricky, the Zephyrus G15 is known for its excellent balance of performance and efficiency. Here's why it's a good fit:

- **Performance:** It typically comes equipped with a powerful AMD Ryzen processor and a dedicated NVIDIA GeForce RTX graphics card, allowing you to play shooting games smoothly.
- **Display:** The 15-inch display is ideal for portability and immersive gaming.
- **Battery Life:** While not exceptional, the Zephyrus G15 is known to offer significantly better battery life than many other gaming laptops in its class, often lasting several hours on a single charge with optimized settings.
- **Price:** Depending on the specific configuration, you should be able to find a model within your 400,000 LKR budget.

Next Steps:

1. **Check Availability and Specific Configurations:** I recommend visiting our website and searching for "ASUS ROG Zephyrus G15." Pay close attention to the processor (Ryzen 7 or Ryzen 9 are ideal) and the graphics card (RTX 3060 or higher is recommended for a good gaming experience).
2. **Compare Prices:** Prices can vary depending on the retailer and the specific configuration.
3. **Read Reviews:** I look for reviews specifically mentioning battery life while gaming or

Type your message here...



The system can filter products by price range and other criteria mentioned by the user.

Scenario 3: Detailed Product Information

When a user asks about specific products, the agent provides detailed information:



Tell me about the ASUS ROG Zephyrus G15



The ASUS ROG Zephyrus G15 is a powerhouse of a gaming laptop packed into a surprisingly thin and light chassis. Here's a breakdown of what makes it a great choice:

Key Features and Benefits:

- **Excellent Performance:** Powered by AMD Ryzen processors (typically Ryzen 7 or Ryzen 9) and NVIDIA GeForce RTX graphics cards (RTX 3060, RTX 3070, or RTX 3080), the Zephyrus G15 delivers exceptional gaming performance, allowing you to play the latest AAA titles at high settings. It also handles demanding tasks like video editing and content creation with ease.
- **Stunning Display:** The 15.6-inch display usually boasts a high refresh rate (often 144Hz or higher) and excellent color accuracy, providing a smooth and immersive gaming experience. Some models even feature a QHD (1440p) resolution for sharper visuals.
- **Impressive Battery Life:** Unlike many gaming laptops, the Zephyrus G15 is known for its relatively long battery life. While you won't get all-day battery life while gaming, it can last for several hours of web browsing, video playback, or other productivity tasks. This makes it a more versatile option for students or professionals who need a laptop that can handle both work and play.
- **Thin and Light Design:** Despite its powerful components, the Zephyrus G15 is surprisingly thin and light, making it easy to carry around. This is a major advantage over bulkier gaming laptops.
- **Quiet Cooling:** ASUS's ROG Intelligent Cooling system keeps the laptop running cool and

Type your message here...



The AI leverages its knowledge of the product catalog to generate informative responses.

Challenges and Solutions

Challenge	Solution	Impact
Contextual Conversations	Implemented chat history tracking to provide conversation context to the AI model	Enabled more coherent multi-turn conversations
Dynamic UI Updates	Used Streamlit's session state to manage application state	Created a responsive interface that updates based on conversation flow

Conclusion

The AI Sales Agent successfully demonstrates the integration of conversational AI with a product recommendation system. The application provides a natural interface for users to explore products, receive recommendations, and make informed purchasing decisions.

By using modular design principles and leveraging the capabilities of the Gemini API, the system delivers an interactive shopping experience that adapts to user inquiries in real-time.